

Executive Summary

Goal: Analyzing the impact of early-stage architectural design decisions on building heating and cooling energy requirements using the UCI Energy Efficiency dataset.

Objectives:

- Identify which building parameters most strongly drive energy performance.
- Evaluate predictive models for reliable HVAC load forecasting.

Key Finding 1: Building Geometry Relative Compactness (ratio of volume to surface area) is the single strongest predictor. Compact forms reduce both Heating and Cooling Loads by minimizing exposed envelope area.

Outcome: Prioritizing compactness is the most effective energy-saving strategy.

Key Finding 2: Height and Glazing Are Critical Drivers

- **Overall Height** significantly increases both heating and cooling energy.
- **Glazing Area** (window-to-floor ratio) above ~**25%** substantially increases loads year-round.

Outcome: Limit unnecessary height and control glazing area to improve efficiency.

Key Finding 3: Orientation Has Negligible Impact Contrary to common assumptions, orientation and glazing distribution show minimal influence on loads in this analysis.

Outcome: Redirect design effort from orientation to geometry and glazing decisions.

Model Test and Results:

- **Linear Models:** Strong performance performance ($R^2 \approx 0.89\text{--}0.91$) but fail statistical assumptions, which makes it not reliable.
- **Random Forest:** Has a near-perfect predictions ($R^2 \approx 0.997$) and extremely low error, which also captures complex nonlinear interactions.

Outcome: Both models perform well but only Random Forest provides reliable results.

Actionable Recommendations

1. **Prioritize compact building forms** to minimize energy loads.
2. **Limit glazing area above 25%** unless mitigated by advanced strategies.
3. **Avoid excessive height** without compensating design features.
4. **Perform ML (Random Forest)** early in design to rapidly evaluate energy implications.
5. **Do not prioritize orientation** in favor of geometry and glazing optimization.

Conclusion

- **Shape is the number one driver** of building energy performance.
- Linear Regression cannot capture all the complex interaction between variables as well as Random Forest.
- Strategic choices in compactness, glazing, and height can significantly reduce energy use, improve sustainability, and lower operational costs.